

The background of the slide is a vibrant blue and cyan gradient. On the left side, there is a stylized illustration of a computer desk with two monitors and a chair. Overlaid on this is a large, semi-transparent shield icon, symbolizing cybersecurity. The overall aesthetic is modern and tech-oriented.

SME Cybersecurity Risk Assessment

BY : ADITYA SONUNE

A practical framework for small and medium enterprises to identify vulnerabilities and prioritize protection without expensive consultants.

Why SMEs Are Prime Targets

Cybersecurity isn't just a "big tech" problem anymore. Small and Medium Enterprises (SMEs) are often the easiest targets because they lack massive IT budgets or dedicated security teams.

This project creates a simple, working way for small businesses to figure out where they're vulnerable without hiring expensive consultants.



The Most Common Threats

During research, it became clear that hackers stick to what works rather than using "movie-style" high-tech tricks:



Phishing

Fake emails tricking people into giving up passwords. The #1 way businesses get hacked because it's easy to do.



Ransomware

All company files get locked up and hackers demand money to release them. The nightmare scenario.



Bad Habits

Weak passwords or employees accidentally deleting sensitive databases due to lack of training.



Old Software

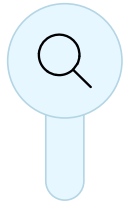
Legacy systems haven't been updated in years. Known holes that hackers scan for automatically.



Building the Model

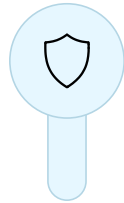
The NIST Cybersecurity Framework is the industry gold standard, but it's way too complicated for small shops. This project strips it down to the basics.

The Five-Step Process



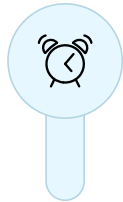
IDENTIFY

Know what you have: servers, laptops, customer data.



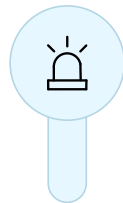
PROTECT

Lock the doors with firewalls and strong passwords.



DETECT

Set up alarms to know if someone is inside your network.



RESPOND

Know exactly what to do the second you find a breach.



RECOVER

Get back to work without losing everything.

The Risk Score Formula

To make this practical, a basic math formula was used:

Risk Score = Likelihood × Impact

A scale of 1 to 5 is used for both factors:

Score	Likelihood	Impact
1	Barely ever (once a year)	Not a big deal
2	Might happen twice a year	Small headache (1hr downtime)
3	Likely once a quarter	Actually hurts (data loss)
4	Happens monthly	Serious trouble (fines/client loss)
5	Happens all the time	Business-ending event

Real Examples Tested

Accountant Phishing Case

A local accounting firm had an email password stolen.

Risk Score: 16 (High) because it's so common.

The Fix: Adding MFA (Multi-Factor Authentication) would have stopped the hacker, even with the right password.

Clinic Ransomware Case

A dental clinic's IT provider got hacked, which spread to the clinic.

Risk Score: 10 (Moderate)

The Fix: They had a "3-2-1" backup (offsite and offline), so they just wiped the PCs and restored everything for free.



The Tools Built

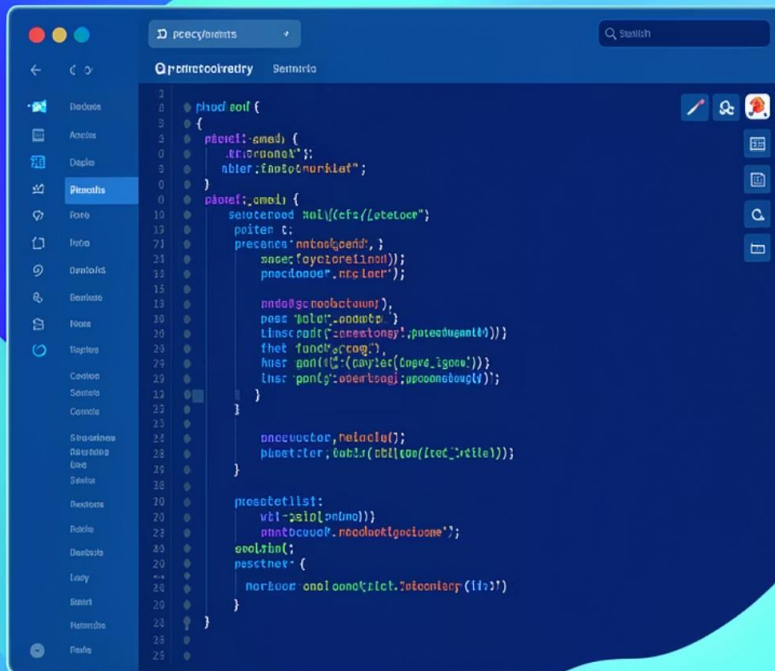
Python Calculator

A script (``risk_calculator.py``) that does the math automatically. Feed it different threats, and it'll tell you if it's "Low", "Moderate", or "High" priority. Creates a CSV file for managers.

Run it with: `python risk_calculator.py`

Web Dashboard

For people who don't like using a terminal, an interactive dashboard (``index.html``) with sliders to see how risk levels change in real-time. Includes Dark Mode.



Top Ways to Stay Safe

Based on research, most small businesses can be "safe enough" if they just do these four things:

- 1** Use MFA everywhere
No exceptions. Multi-Factor Authentication stops most attacks even if passwords are compromised.
- 2** The 3-2-1 Backup
Keep 3 copies, use 2 media types, and store 1 copy offsite.
- 3** Patch your stuff
Set everything to "Auto-Update" to close known security holes.
- 4** Train your team
Most hacks start with human error, not computer error. Regular training is essential.

Key Takeaway

You can't stop every hack. You have to assume you'll get hit eventually.

The goal of this project was to show that by being smart with a simple risk matrix and basic tools, even a tiny business can be tougher to hack than a big bank.



THANK YOU

